A centre whose orbit meets the discontinuities lies in a non-trivial connection

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Editorial note. The counted statement is the author's labelled lemma environment at source0.tex lines 652-670; its proof environment opens at line 672, the line after the blank separator, so the pairing is by adjacency and the retained proof range 672-682 is the whole environment.

Editorial note. Four numbered environments are retained as context because the statement or the proof refers to them: eq:sym_identity_endpoints_2 (source lines 480-482), eq:sym_identity_endpoints (source lines 475-478), eq:inverse_iterates (source lines 533-537) and eq:delta_1/2 (source lines 539-544).

Editorial note. This source is new to the material-authoring work and its preamble is built from the author's own macro and theorem block (source lines 8-45) rather than from a prepared adapter.

Editorial note. This source has no label-number dump and the retained counter values are left empty, so the printed numbering of the card is the wrapper's default; the identity of the counted item is its author label.

Editorial scope. One original result of the pinned publisher article, counted once: a centre whose orbit meets the discontinuities lies in a non-trivial connection (author label lem:symmetric_connection).

Allowed standard prerequisites (editorial).

- Symmetry and the conjugacy equation. The reflection of the interval conjugates the transformation to its inverse off the discontinuities.. It is the hypothesis and the source of the endpoint identities.
- Centre of a continuity interval. c_beta is the middle point of the continuity interval indexed by beta, and c_{1/2} is the middle of the whole interval.. It is the point whose orbit is followed.
- Connection. A connection is a finite orbit fragment joining two discontinuities.. It is the structure that the conclusion places the centre in.

$$T^{-1} \circ \mathcal{I}_I(\partial I_\alpha) = \partial I_{\hat{\alpha}}, \text{ where } \pi_0(\hat{\alpha}) = \pi_0(\alpha) - 1, \tag{1}$$

$$\mathcal{I}_I \circ T(\partial I_\alpha) = \partial I_{\hat{\alpha}}, \text{ where } \pi_0(\hat{\alpha}) = \pi_0(\alpha) + 1, \tag{2}$$

$$T^{-m}(c_\alpha) = T^{-1} \circ \mathcal{I}_I\big(T^{m-\delta_{1/2}(\alpha)}(c_\alpha)\big), \tag{3}$$

$$\delta_{1/2}(\beta) = \begin{cases} 1 & \text{if} \quad \beta = 1/2, \\ 0 & \text{otherwise.} \end{cases} \tag{4}$$

Lemma 0.1. *Let $T : I \to I$ be a symmetric IET and $\beta \in \mathcal{A} \cup \{1/2\}$. Assume there exists $m \in \mathbb{Z}$ and $\alpha \in \mathcal{A}$ such that $T^m(c_\beta) = \partial I_\alpha$ and $T^k(c_\beta) \notin \{\partial I_\gamma\}_{\gamma\in\mathcal{A}}$, for any $-|m| < k < |m|$.*

Then if $m \geq 0$, we have $\alpha \neq \pi_0^{-1}(1)$ and

$$T^{-m-\delta_{1/2}(\beta)}(c_\beta) = \partial I_{\hat{\alpha}},$$

where $\delta_{1/2}$ is given by (4) and $\pi_0(\hat{\alpha}) = \pi_0(\alpha) - 1$.

If on the other hand $m < 0$, then $\alpha \neq \pi_1^{-1}(1)$ and

$$T^{-m-\delta_{1/2}(\beta)}(c_\beta) = \partial I_{\hat{\alpha}},$$

with $\pi_0(\hat{\alpha}) = \pi_0(\alpha) + 1$.

In particular, c_β lies inside a non-trivial connection.

Proof. If $m = 0$, then necessarily $\beta = 1/2$ and we have $c_{1/2} = \partial I_\alpha$ with $\pi_1(\alpha) \neq 1$. By (1), $T^{-1}(c_{1/2}) = \partial I_{\hat{\alpha}}$, where $\pi_0(\hat{\alpha}) = \pi_0(\alpha) - 1$.

Without loss of generality, let us assume $m < 0$, the case $m > 0$ being analogous. By (3),

$$T^m(c_\beta) = T^{-1} \circ \mathcal{I}_I(T^{-m-\delta_{1/2}(\beta)}) \iff T^{-m-\delta_{1/2}(\beta)}(c_\beta) = \mathcal{I}_I \circ T\big(T^m(c_\beta)\big).$$

Notice that since $T^{m+1}(c_\beta) \notin \{\partial I_\gamma\}_{\gamma\in\mathcal{A}}$ then $T^m(c_\beta) = \partial I_\alpha \neq \partial I_{\pi_0^{-1}(1)}$. Hence, by (2), the equation above implies $T^{-m-\delta_{1/2}(\beta)}(c_\beta) = \partial I_{\hat{\alpha}}$, where $\pi_0(\hat{\alpha}) = \pi_0(\alpha) + 1$. □